

SAHIL YADAV

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SUMMARY

Applied mathematician and ML engineer specializing in **large-scale optimization and decision systems under uncertainty** – stochastic optimization, demand forecasting, deep learning and ads. Track record of shipping models with direct, multi-crore P&L impact on messy operational systems; published research in deep learning and GANs.

EDUCATION

Birla Institute of Technology and Science, Pilani

2018 – 2023

M.Sc. (Hons.) Mathematics + B.E. (Hons.) Electrical & Electronics Engineering

WORK EXPERIENCE

Glance, Bengaluru

Aug 2026 – Present

Applied Scientist 3

· Working on **ad auction floor-price prediction** using **XGBoost & contextual-bandit models** across major OEMs, targeting higher revenue with a game theory approach to understand DSP behavior.

Zepto, Bengaluru

Jul 2023 – Aug 2026

Data Scientist I → II → Senior Data Scientist

· Engineered a large-scale **demand forecasting engine** with custom loss functions, delivering a **5% MAPE / 10% Bias reduction** and a **₹ 22 Cr reduction** in fortnightly procurement.

· Designed an end-to-end **warehouse-store vehicle scheduling & routing model** with MILP optimization under OSRM, capacity, manpower & timing constraints; **INR 5 Cr monthly** cost reduction nationwide.

· Led cross-functional development of Zepto's **Auto-Replenishment System** – an omnichannel transfers generator, yielding a **10% availability uplift** and faster forecast-to-execution.

· Built a **pick-path optimization system** via co-occurrence-based placement and affinity-driven sequencing, cutting **average travel distance ~30%**.

· Guided development of an **assortment recommendation model** fusing user-search signals, relevance scoring & competitor data to surface emerging demand.

Zepto, Mumbai — *Data Science Intern*

Summer 2022

· Formulated a **PID control-based dynamic pricing model** balancing real-time demand against supply stress while minimizing order loss.

Pixis AI, Bengaluru — *Data Science Intern*

Fall 2021

· Built a **relevance ranking model** over video metadata using embedding similarity and NLP to improve ad conversion.

Google Summer of Code, OpenAstronomy — *Contributor*

Summer 2020

· Contributed to **CTLearn**: trained RNN-based models for gamma-ray event reconstruction and classification, and integrated ROOT data pipelines for MAGIC and VERITAS telescope data.

OPEN-SOURCE PROJECT

Scout — *Personal project*

Live app · GitHub

Built a grocery intelligence app to help Indians make smarter choices. Integrated **DeepSeek** for ingredients rating/tagging, goal-based ranking and free-form querying.

PUBLICATIONS

S. Yadav, A. Mehra, H. Rohmetra, R. Ratnakumar, P. Narang. **DerainGAN: Single Image Deraining Using Wasserstein GAN**. *Multimedia Tools and Applications*, Springer.

T. Miener, R. López-Coto, ..., S. Yadav. **IACT event analysis with the MAGIC telescopes using deep convolutional neural networks with CTLearn**. *ADASS XXXI*.